

Network-Aware Post-Timetabling Examination Room Allocation with Screened-Pool Repair

Hieu Le Minh¹, Phung Trung Nguyen Nguyen¹, Thuan Mai Duc¹, Ngoc Khanh Tran Nguyen², and Xuan-Bach Le¹*

¹ Faculty of Computer Science and Engineering, Ho Chi Minh City University of Technology, VNU-HCM, Ho Chi Minh City, Viet Nam
`{hieu.lehieu@hks,nguyen.nguyen938,thuan.maiduc2411,lexuanbach}@hcmut.edu.vn`

² University of Science, Viet Nam National University Ho Chi Minh City, Ho Chi Minh City, Viet Nam
`22120378@student.hcmus.edu.vn`

Abstract. Universities often fix the examination timetable before finalizing room allocation. We start from this fixed timetable and an incumbent room allocation; examination sessions are not rescheduled. We study the downstream post-timetabling examination-room repair problem on real anonymized data from a large public university, covering 81 examination sessions over 19 days, 115,641 candidates, and 3,392 incumbent room assignments. The incumbent plan uses 38.1% of active capacity and has 39.4% mean-room physical utilization. We formulate the task as bipartite network repair with two operational regimes: *Legacy-room repair*, which reassigns packages among incumbent active rooms, and *Screened-pool repair*, which expands the candidate-room set with a small controlled pool of unused rooms. We compare a weighted ILP-based policy with transparent repair heuristics. The ILP-family policy reaches 57.6% room reduction under Legacy-room repair and 62.1% under Screened-pool repair, with solver status reported explicitly. Screened greedy reaches 45.3%, robust repair trades savings for higher stress-test feasibility, and adaptive repair recovers 41.8% reduction with millisecond-scale median runtime. We complement the deterministic institutional evaluation with controlled attendance-buffer and room-loss stress tests, which assess feasibility under perturbed assumptions rather than probabilistic attendance forecasts, and with a synthetic-room public-benchmark cross-check. The contribution is an applied, operational repair framework and artifact-driven evaluation for fixed-timetable examination room allocation, rather than a new general-purpose optimization algorithm.

Keywords: Examination room allocation · Educational timetabling · Integer linear programming · Greedy matheuristics · Computational networks

* Corresponding author.

1 Introduction

Unlike from-scratch examination timetabling, we study the fixed-timetable room layer: sessions are not rescheduled, while administrators choose active rooms, safe package moves, and capacity slack for late changes. Active rooms carry invigilation, setup, wayfinding, monitoring, and contingency costs.

We view the task as network repair. Each session is a bipartite package–room graph whose edges encode capacity, campus, allowed-room, and compatibility policies. Repair rewires this graph to close rooms by consolidating incumbent active rooms or adding screened unused rooms.

Our motivating dataset comes from a large public university in Vietnam: 81 examination sessions over 19 days, 115,641 candidates, and 3,392 baseline room assignments. The incumbent uses 38.1% of active capacity and 39.4% mean-room physical utilization. *Legacy-room repair* uses incumbent active rooms, whereas *Screened-pool repair* augments them with $P_t(k)$, with $k = 3$ as the main setting. This tests controlled expansion rather than from-scratch reassignment.

For both regimes we use a weighted HiGHS ILP and transparent baselines. The weighted scalar objective gives active-room count a dominant primary weight; active-capacity slack and incumbent disruption are lower-weight secondary preferences, so the implementation is a weighted scalarization, not a multi-stage procedure. Greedy and local repair expose operational moves, while robust and adaptive variants test controlled attendance-buffer and room-loss scenarios. The main evaluation reports 62.1% reduction for screened-pool ILP, 45.3% for screened greedy, 27.0% for robust repair, and 41.8% for adaptive repair.

Prior work shows that exact/greedy hybrids can save rooms on real data [15], while classroom-assignment models provide foundations [10,25]. We build on educational-timetabling benchmarks and surveys [7,22,19,20,8,24], but isolate post-timetabling repair and make edge insertion explicit.

The paper makes four applied contributions:

- We formulate the fixed-timetable, post-timetabling examination-room allocation task as bipartite network repair with two operational regimes: Legacy-room repair and Screened-pool repair.
- We implement a HiGHS-backed weighted scalar ILP with a dominant active-room term, together with greedy and randomized baselines, local repair, model-reduction repair, robust repair, and adaptive repair in one reproducible harness.
- We evaluate the framework on a real anonymized 81-session institutional dataset, reporting room reduction, active utilization, physical utilization, runtime, solver status, and stress-test feasibility.
- We provide a reviewer-auditable evaluation protocol: a full attendance-buffer and room-loss stress grid labeled as a stress test rather than a calibrated forecast, plus a synthetic-room public cross-check that is kept separate from the deployment evidence.

Evidence comes from one institution; the public conversion lacks real room pools; and adaptive repair is an operator-selection procedure, not a learned optimizer. Thus Legacy-room repair explains most deterministic savings, while Screened-pool repair adds an ILP-family room-reduction gain.

2 Related Work

Educational timetabling. Examination timetabling is a mature area with conflicts, capacities, and institutional constraints [7,22,16]. The ITC2007 exam track established comparison norms [19,20], and surveys cover exact, heuristic, metaheuristic, and hyper-heuristic approaches [8,24]. Our focus is the remaining room-layer repair once periods are fixed.

Post-timetabling room allocation. The closest work studies room assignment after scheduling. Dammak et al. [10] use 0–1 programming and transportation variants; Soghier and Qu [25] combine graph-colouring and bin-packing; and Lemos et al. [15] study two-stage ILP with greedy and GRASP. We build on these approaches while separating Legacy-room repair, which reuses incumbent rooms, from Screened-pool repair, which augments the candidate set.

ILP formulations and case studies. Mixed-integer programming has been used for examination models [4], course timetabling [5], exam case studies [13], and graph-based university timetabling [11,23]. These studies support ILP for moderate structured instances; our split keeps augmentation in a small screened pool.

Matheuristics and repair. Hybrid search is well established [6,3], and fix-and-optimize or quality-recovery matheuristics combine construction, decomposition, and local exact improvement [17,18,21]. Our repair remains transparent: Legacy-room repair uses decreasing-size packing and post-pass closing, Screened-pool repair adds a gain-based phase, and ILP rows coordinate difficult consolidations.

Robust and stochastic timetabling. Robust examination timetabling models uncertain data and quality–robustness tradeoffs [1,2], while stochastic capacitated timetabling introduces two-stage uncertainty and recourse [9]. This motivates our stress grid, not a calibrated robust optimization model.

Gap addressed. Our gap is operational: after the timetable is fixed, room repair must be strong, explainable, and policy-bounded. We combine a two-regime formulation, reproducible HiGHS ILP, transparent baselines, stress testing, and a real large-university study. The contribution is an applied network-aware protocol, not a new general-purpose optimization algorithm.

3 Problem Formulation

Because the timetable fixes all exam times, we optimize each examination session independently. Let T denote the set of sessions. For each session $t \in T$ we

are given a set of *packages* (or *groups*) G_t , a set of *rooms* R_t , the incumbent assignment $b_{g,r} \in \{0, 1\}$, package sizes n_g , room capacities c_r , and package–room feasibility $a_{g,r} \in \{0, 1\}$ (encoding room features, campus separation, and exam-policy restrictions), and package–package compatibility $m_{g,h} \in \{0, 1\}$. We set $m_{g,h} = 1$ exactly when g and h may share a room; in the institutional rule used here, two packages may share a room only if their subjects differ.

The network object is a bipartite graph $H_t = (G_t \cup R_t, E_t)$ with an edge $(g, r) \in E_t$ iff $a_{g,r} = 1$. A solution selects edges so that every package is covered exactly once, every active room respects capacity, and incompatible packages are not co-located. The formulation therefore treats room allocation as graph repair: the incumbent graph is feasible, and the optimizer rewires it to close rooms.

Decision variables. $x_{g,r} \in \{0, 1\}$ indicates assignment of package g to room r ; $y_r \in \{0, 1\}$ indicates that room r is active.

Common constraints. For each session t ,

$$\sum_{r \in R_t} x_{g,r} = 1 \quad \forall g \in G_t, \quad (1)$$

$$\sum_{g \in G_t} n_g x_{g,r} \leq c_r y_r \quad \forall r \in R_t, \quad (2)$$

$$x_{g,r} \leq a_{g,r} \quad \forall g \in G_t, r \in R_t, \quad (3)$$

$$x_{g,r} + x_{h,r} \leq 1 + m_{g,h} \quad \forall g \neq h, r \in R_t. \quad (4)$$

Constraint (1) assigns each package once, (2) couples capacity with room activation, (3) removes infeasible package–room edges, and (4) forbids two packages with the same subject from sharing a room.

Objective. The institution prioritizes reducing active-room count, followed by lower-priority considerations of active capacity and disruption from the incumbent allocation. We encode these preferences using the fixed weighted scalar objective in (5):

$$\min_{x,y} \quad \alpha \sum_{r \in R_t} y_r + \beta \sum_{r \in R_t} \left(c_r y_r - \sum_{g \in G_t} n_g x_{g,r} \right) + \gamma D(x, b), \quad (5)$$

where $D(x, b) = \sum_{g,r} |x_{g,r} - b_{g,r}|$ counts changed package–room assignments and $\alpha \gg \beta \gg \gamma > 0$. The weighting scheme therefore represents a strong practical priority among the three criteria; we describe the implemented formulation as a weighted scalarization rather than as a multi-stage optimization procedure.

Two operational regimes. The institutional setting separates cleanly into two regimes that differ only in the admissible room set:

- **Case 1 (Legacy-room repair).** The admissible rooms are exactly those used by the incumbent: $R_t^{(1)} = \{r : \exists g, b_{g,r} = 1\}$. The model may consolidate package assignments only among incumbent active rooms; room identities remain unchanged.

- **Case 2 (Screened-pool repair).** The admissible rooms are $R_t^{(2)} = R_t^{(1)} \cup P_t(k)$, where $P_t(k)$ is a small predefined pool of currently unused rooms screened for the session. The model may activate rooms from $P_t(k)$ only when they help close incumbent rooms. In the main evaluation we use $k = 3$ and retain converted $k \in \{0, 1, 5\}$ regimes for follow-up sweeps.

The two regimes use the same constraint set (1)–(4) and objective (5). They differ only in which room nodes may become active. Legacy-room repair measures what reorganization among incumbent active rooms can achieve. Screened-pool repair measures the value of controlled candidate-set expansion. This distinction is the technical hinge of the paper.

Stress-test scope. Beyond the deterministic problem, Sections 6 and 7 report a controlled stress-test grid that multiplies demand by an attendance-buffer factor ρ and removes a fraction δ of rooms from R_t to model late unavailability. The reported grid uses all nine combinations $\rho \in \{1.00, 1.05, 1.10\}$ and $\delta \in \{0, 0.05, 0.10\}$. These perturbations are simulated from session-size and capacity statistics. We therefore use them as controlled stress tests, not calibrated probabilistic forecasts or predictions of future events.

4 Methods

We solve Section 3 with a weighted scalar ILP and repair heuristics. Methods operate session by session because the timetable is fixed and rooms are uncoupled; this matches administrative inspection of activations, reassignments, and closures.

4.1 ILP Solved by Off-the-Shelf Solvers

For each session we instantiate (1)–(5) with binary $x_{g,r}$ and y_r and pass it to HiGHS. Legacy-room repair restricts r to incumbent rooms; Screened-pool repair adds the top- k unused rooms by capacity within the compatible campus partition. The binary count is $O(|G_t| \cdot |R_t^{(\cdot)}|)$ and remains moderate because packages and the screened room set are small. Sessions above the exact threshold are marked skipped-large rather than claimed optimal.

The weighted scalar objective gives active-room count the dominant primary weight; active-capacity slack and incumbent disruption are lower-weight secondary preferences. We therefore describe it as a weighted scalarization, not a multi-stage procedure. The artifact records solver status, objective value, bound, gap, and termination for each ILP-family row.

4.2 Greedy and Repair Heuristics

The greedy method has two phases that correspond to the two operational regimes. Phase 1 handles Legacy-room repair; Phase 2 extends Phase 1’s output to Screened-pool repair by considering rooms from $P_t(k)$.

The procedure is deliberately simple. Phase 1 sorts packages by descending size and assigns each package to the feasible active room with least residual capacity; if none is feasible, it opens the package’s incumbent room. A post-pass closes light active rooms when all packages can move elsewhere. Phase 2, used only in Case 2, scans P_t for a room that can absorb at least two compatible packages. The accepted move maximizes room-count gain and breaks ties by lower slack. All moves are checked against capacity and subject-disjointness constraints.

Greedy variants. Beyond deterministic greedy, the harness evaluates largest-first and randomized variants using the same session information, candidate rooms, and feasibility checks. They provide low-runtime comparison points; no optimality guarantee is claimed.

Local and model-reduction repair. Fixed local search and model-reduction ILP provide bounded improvement or polishing around the checked repair; no general optimality guarantee is claimed for the neighborhood heuristic.

Robust and adaptive repair. Robust greedy uses demand and capacity buffers. Adaptive repair selects among close-room repair, robust construction, and model-reduction polishing using observed improvement. These are transparent operators, not a learned optimizer.

Complexity and runtime profile. Each Legacy-room pass is $O(|G_t| \cdot |\mathcal{B}|)$, and the post-pass runs at most five times. The Screened-pool phase considers $|P_t|$ candidate rooms in a single accept-best loop. In practice $|G_t|, |R_t|, |P_t|$ are at most a few hundred per session, so the greedy method runs in milliseconds to a second per session.

Why pair greedy with ILP. The ILP-based policy is a solver reference subject to the reported statuses. Greedy is the deployment-oriented counterpart: it updates quickly as exam-day data changes, records moves as “activate room”, “re-assign package”, or “close room”, and permits administrator overrides. Section 6 measures the quality–runtime trade-off.

5 Experimental Setup

The evaluation has three roles: deterministic institutional results, controlled attendance-buffer and room-loss stress tests, and an auxiliary synthetic-room cross-check. Artifacts are inspectable per session, day, regime, and solver so that the source of each saving is clear.

Primary dataset. The primary dataset is an anonymized extract from a large public university in Vietnam: 81 sessions over 19 days, 115,641 candidates, and 3,392 baseline room assignments. The incumbent has 38.1% active-capacity and 39.4% mean-room physical utilization. Each session provides package identifiers and sizes, incumbent rooms, capacities, and package–room eligibility filters. The screened pool $P_t(k)$ uses currently unused compatible rooms, with $k = 3$ in the main evaluation.

Auxiliary public stress test. To test whether greedy-ILP ordering is institution-specific, we report a structural cross-check on synthetic-room conversions of a classic public exam benchmark family [7,22,26]. Because these benchmarks lack room capacities, we generate room pools and label results *synthetic-room stress tests*; they test conflict shapes and demand scales, not deployment claims.

Solvers and harness. We use HiGHS 1.14.0 as the open-source MIP backend [12]. The Python harness builds the common schema, applies a 30 s per-session limit, and records status, objective, bound, gap, and termination. The same fixed weighted objective and machine are used throughout.

Methods compared.

- INCUMBENT: the operational room assignment as deployed by the institution.
- LEGACY GREEDY/LEGACY ILP: Legacy-room repair using incumbent active rooms only.
- SCREENED GREEDY/SCREENED ILP: Screened-pool repair with $k = 3$.
- ROBUST REPAIR: conservative greedy construction under generated demand and capacity buffers.
- ADAPTIVE REPAIR: local repair with model-reduction polishing when the session is below the exact threshold.

Research questions. The evaluation addresses four questions on the operational data and one auxiliary cross-check.

RQ1 How much room can be saved over the incumbent in each regime, in aggregate and per day?

RQ2 What is the value of Screened-pool repair relative to Legacy-room repair?

RQ3 How do deterministic, robust, and adaptive repair baselines trade room savings against stress-test feasibility?

RQ4 How do screened-pool solutions behave under attendance-buffer and room-outage stress-test configurations?

RQ5 (auxiliary) Does the same qualitative ordering hold on the synthetic-room conversions of the classic public exam family?

Metrics. We report active-room count, room reduction, active and physical utilization, disruption $D(x, b)$, ILP objective/gap, wall-clock time, termination status, mean stress overflow, and feasibility without recomputation.

Reporting protocol. For RQ1–RQ3 we report aggregate summaries and retain per-session artifact tables; paired ILP–heuristic comparisons use checked repairs. For RQ4 we report the 3×3 grid as a controlled stress test because no historical attendance or outage records exist, so its percentages are not probabilities.

Reproducibility. The package includes anonymized instances, screened-pool/model builders, solver configs, repair baselines, sensitivity scripts, aggregate tables, and public-test conversion scripts; sensitivity seeds are fixed.

Table 1. Main results on the real institutional dataset. Red. denotes room reduction relative to the incumbent; Act. util. denotes active-room utilization (demand divided by active room capacity); Phys. util. denotes mean physical utilization (mean per-room demand/capacity ratio); Mean t. and Max t. denote mean and maximum per-session runtimes. Arrows mark preferred direction.

Method	Red.↑	Act. util.↑	Phys. util.↑	Mean t.↓	Max t.↓
Incumbent	0.0%	38.1%	39.4%	0.0004 s	0.0018 s
Legacy Greedy	45.3%	85.2%	85.4%	0.0004 s	0.0017 s
Legacy ILP	57.6%	91.8%	91.5%	5.91 s	30.52 s
Screened Greedy	45.3%	85.3%	85.4%	0.0004 s	0.0018 s
Screened ILP	62.1%	93.4%	93.3%	5.59 s	30.53 s
Robust Repair	27.0%	59.6%	60.0%	0.0005 s	0.0018 s
Adaptive Repair	41.8%	77.4%	77.9%	0.04 s	2.50 s

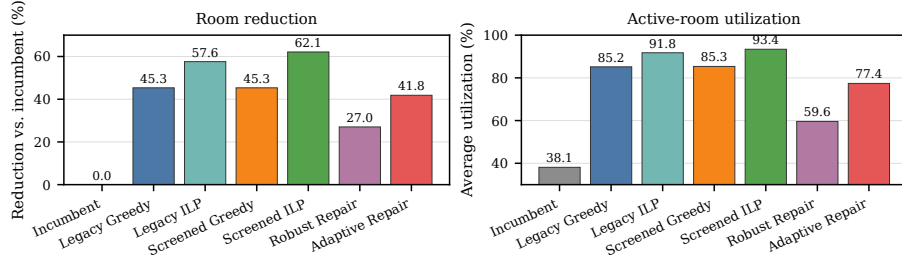


Fig. 1. Room savings and active-utilization tradeoff on the real institutional dataset. The points come from the artifact CSV used to generate Table 1.

6 Results

We report institutional results, controlled stress scenarios, and a bounded synthetic-room cross-check. Legacy-room repair uses incumbent active rooms; Screened-pool repair augments the candidate set with screened rooms.

6.1 Deterministic Network-Repair Results

To address RQ1 and RQ2, Table 1 reports the generated aggregate values for 81 sessions. We separate active/effective utilization from mean-room physical utilization: the former is a capacity-weighted active-room ratio, while the latter is the mean per-room demand/capacity ratio. The two quantities are not interchangeable.

ILP-family status note. Solver status is reported explicitly for ILP-family rows, including optimal, time-limit feasible, and skipped-large cases; these aggregates are policy-level ILP-family outcomes, not all-optimal ILP solves.

For RQ1, screened-pool ILP reaches 62.1% room reduction and adaptive repair 41.8%. For RQ2, Screened Greedy remains at 45.3%, so candidate-set

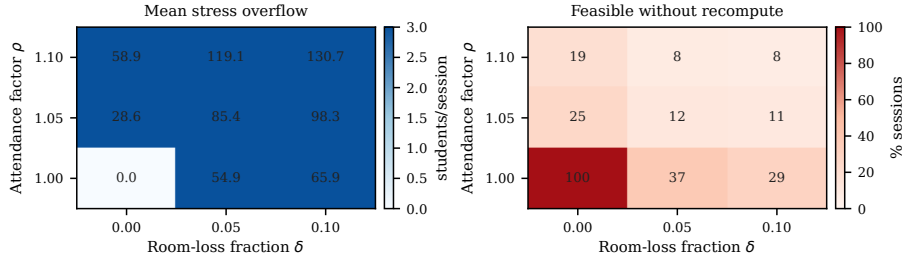


Fig. 2. Controlled sensitivity stress test on Screened-pool adaptive repair: stress test, not forecast. The left panel reports mean stress overflow per session; the right panel reports the fraction of sessions feasible without recomputation.

expansion mainly benefits the ILP-family policy. Table 1 gives the deterministic RQ3 quality–runtime comparison; stress feasibility follows.

Pool-size configuration. The converter materializes screened-pool regimes for $k \in \{0, 1, 3, 5\}$, while the main evaluation executes the Legacy-room repair regime and default $k = 3$ Screened-pool repair regime. We use $k = 3$ because it changes the feasible neighborhood without turning the task into from-scratch reassignment.

Solver comparison. HiGHS is the default open-source ILP backend. The artifact table records solver status, gap, bound, and termination for each ILP-family row, and marks sessions above the polishing threshold explicitly.

6.2 Controlled Sensitivity Stress Test

For RQ3 and RQ4, Figure 2 reports Screened-pool adaptive repair over the nine combinations of $\rho \in \{1.00, 1.05, 1.10\}$ and $\delta \in \{0, 0.05, 0.10\}$. This controlled perturbation is not a calibrated forecast; its percentages are not probabilities of future events.

Averaged over the controlled stress-test grid, adaptive repair attains 27.5% stress-test feasibility. This value summarizes the predefined stress scenarios and should not be interpreted as a forecast probability of future attendance or room availability.

6.3 Auxiliary Synthetic-Room Public Benchmark Cross-Check

To address auxiliary RQ5, we use the Toronto/Carter archive hosted by the University of Nottingham as a public benchmark cross-check with synthetic rooms [7, 22, 26]. The auxiliary conversion parses `car-f-92` and `car-s-91`, which provide exam sizes and student conflicts but not rooms or a fixed timetable. With seed 20260515 and `max_toronto_exams=200`, the converter keeps the 200 largest exams from each instance, builds conflicts from shared students, greedily assigns

Table 2. Auxiliary Carter-style results with generated room pools: synthetic-room stress test; not direct room-allocation evidence. Arrows mark preferred direction. Reduction is room reduction, Active util. is active utilization, Robust feas. is robust feasibility, Mean overflow is mean stress overflow, and Status records F=feasible, R=repaired incumbent, O=HiGHS optimal.

Method	Sessions	Reduction $\uparrow$	Active util. $\uparrow$	Robust feas. $\uparrow$	Mean overflow $\downarrow$	Time $\downarrow$	Status
Incumbent	64	1.1%	73.5%	20.6%	135.5	0.0000 s	16F/48R
Greedy	64	1.1%	73.5%	19.6%	135.9	0.0000 s	64F
ILP	64	26.2%	87.2%	23.2%	153.0	0.02 s	64O
Robust Repair	64	0.0%	72.4%	33.3%	130.3	0.0001 s	64F
Adaptive Repair	64	19.0%	83.5%	24.4%	145.9	0.01 s	43F/21O

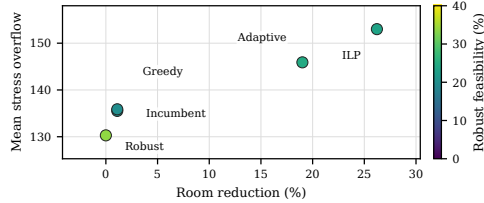


Fig. 3. Room-saving/mean-overflow trade-off with generated room pools: synthetic-room stress test; not direct room-allocation evidence. Color encodes robust feasibility.

conflict-free periods, and emits one fixed-session instance per period. This yields 64 sessions, 400 events, and 78,225 total demand.

Because the files lack real rooms, each period receives a synthetic feasible room per event, sized at 135% of demand and rounded up to the next 10 seats. A seeded shuffle creates incumbent rooms, after which all methods solve the same fixed-period task. These rows test ordering under different conflict shapes and demand scales, but they remain *synthetic-room stress tests*; we exclude them from headline deployment claims.

Table 2 and Fig. 3 address auxiliary RQ5 by showing the same qualitative ordering: ILP saves more generated rooms, robust repair keeps more slack, and adaptive repair sits between them. This cross-check is structural evidence, not direct institutional room-allocation evidence.

7 Analysis

Where the savings come from. For RQ1, the 45–62% reductions reflect low incumbent utilization (38.1% active and 39.4% physical) and sufficient allowed-room slack for consolidation.

What the screened pool changes. For RQ2, Screened-pool repair is not from-scratch reassignment: it adds controlled candidate nodes that may absorb packages while incumbent rooms close. Greedy captures obvious consolidations, ILP coordinates smaller moves, and robust repair sacrifices savings to preserve slack.

Solver status and large sessions. For RQ3, most medium sessions solve within the 30 s limit; larger or time-limited cases are reported by status. ILP generally consolidates more, while heuristics provide faster feasible adjustments, motivating greedy by default and ILP for structurally difficult cases.

What the sensitivity sweep does and does not show. For RQ4, the sensitivity sweep is a controlled stress test, not a robustness claim or attendance forecast: tight packing loses feasibility under compound perturbations, while robust repair preserves slack at lower savings. An explicit utilization margin and screened pool P_t support late recomputation.

Network view and generality. The group–room graph H_t is the structural object: savings come from compatible-package density subject to capacity. The two regimes separate incumbent-edge reuse from controlled expansion. The pattern may recur after fixed scheduling, but its magnitude depends on local utilization, slack, and room inventory.

Recommendation. For similar utilization profiles, use transparent greedy by default, run screened-pool HiGHS ILP when extra reduction is worth solver time, and switch to robust repair when uncertainty matters more than maximum savings.

8 Limitations and Operational Considerations

Results come from one university. The 62.1% reduction depends on incumbent slack/utilization and candidate-room inventory; these conditions are not universal. The synthetic-room cross-check supports method ordering, not direct public room evidence.

The timetable and session assignments remain fixed; timetable construction and rescheduling are outside scope. The sensitivity sweep is a controlled stress test, not a calibrated forecast, and we do not claim real-world robust feasibility. Adaptive repair is transparent operator selection, not a learned optimizer; learning-augmented ranking remains future work.

The private dataset permits release only of anonymized derived instances, schemas, generators, validators, and aggregate tables. Some ILP rows are time-limit or skipped-large rather than proven optimal; status fields and aggregates should be read accordingly. Fairness, spacing, accessibility, and invigilator workload remain hard constraints or administrator-reviewed policies.

9 Conclusion

We show that post-timetabling examination-room allocation is applied bipartite network repair with packing constraints. On 81 sessions, Screened-pool ILP reduces active rooms by 62.1%, Screened Greedy by 45.3%, Robust Repair by 27.0%, and Adaptive Repair by 41.8%. Controlled candidate expansion adds an ILP-family gain beyond Legacy-room repair, while most savings come from

incumbent reassignment. We provide a network-aware operational formulation and reproducible quality–runtime evaluation, not a fundamentally new general-purpose optimization algorithm.

Acknowledgments. We acknowledge Ho Chi Minh City University of Technology (HCMUT), VNU-HCM for supporting this study.

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